<table>
<tr><td>12</td><td>Answer: D</td></tr>
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Question asked for “rate of heat loss”. [J s^{-1}]

- Eliminate A and C by unit analysis.

  Note that m is defined as rate of loss of mass. [kg s^{-1}]

  $$\left[\frac{P}{m}\right]=\left[\frac{E/t}{m}\right]=\frac{\text{J s}^{-1}}{\text{kg s}^{-1}}=\frac{\text{J}}{\text{kg}}$$

- Option B: $P_1 - P_2$ removes the rate of heat loss term. Eliminate.

- Mathematical proof for D:

$$P_1 m_2 - P_2 m_1 = \frac{\cancel{m_2 m_1 l_v} + m_2 h - \cancel{m_1 m_2 l_v} - m_1 h}{m_2 - m_1} = \frac{\cancel{(m_2 - m_1)} h}{\cancel{m_2 - m_1}}$$

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